INCH-POUND
MIL-DTL-49285/2B
w/AMENDMENT 1
18 November 2021
SUPERSEDING
MIL-DTL-49285/2B
2 March 2018

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, FIFTEEN INDIVIDUALLY SHIELDED PAIRS OF 22 (7 X 30) AWG

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

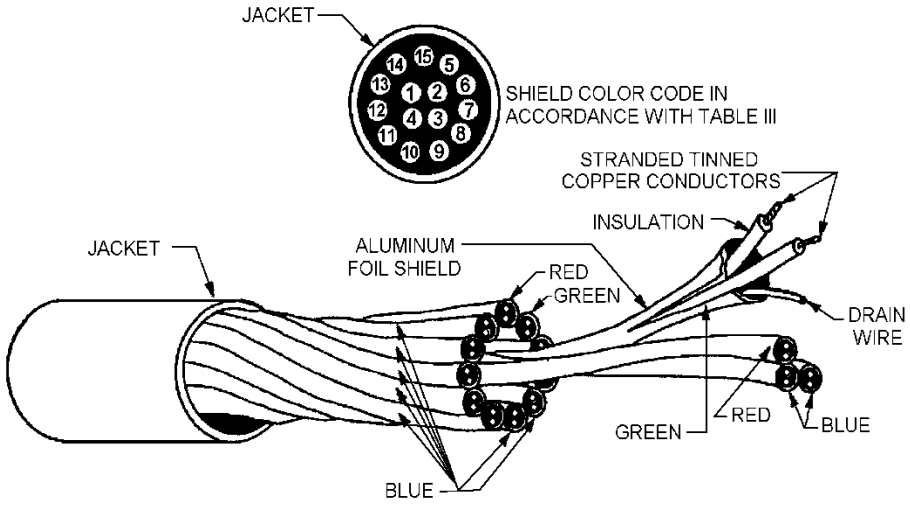



FIGURE 1. Cable illustration.

Engineering information: See table I.

Note: Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.



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TABLE I. Description, electrical.

| Electrical characteristics | Requirements |
|---|---|
| Nominal capacitance between conductors | 30 pF/ft (picofarads/foot) |
| Nominal capacitance between the conductor and the shield with one conductor tied to the shield | 55 pF/ft |
| Continuous working voltage | 300 $V_{rms}$ maximum between conductors |
| Nominal capacitance between adjacent shields | 150 pF/ft |
| Continuous working voltage between adjacent shields | 50 volts maximum |
| Shield dc resistance | 14 ohms/1,000 feet maximum |

TABLE II. Description, physical.

| Components | Construction details |
|---|---|
| Number of pairs | Fifteen individually shielded |
| Conductor type and wire size | Stranded tinned copper 22 AWG |
| Conductor stranding | 7 X 30 AWG |
| Conductor insulation | Polypropylene |
| Conductor insulation thickness | 0.010 inch nominal thickness |
| Drain wire type and size | Stranded tinned copper 22 AWG |
| Drain wire stranding | 16 X 34 AWG |
| Jacket material | Polyvinyl chloride |
| Jacket thickness | 0.050 inch |
| Finished cable diameter | 0.600 inch maximum outside diameter |
| Cable style (UL) | 2493 |
| Tensile strength (jacket) | 2,000 pounds per square inch minimum |
| Elongation (jacket) | 150 percent minimum |
| Overall cabling lay length | 1.3 twists per foot ± 10 percent |

Cabled pair orientation: The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).



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TABLE III. Shield color code.

| Pair numbers | Shield color |
|---|---|
| Center core | |
| 1st pair | Red |
| 2nd pair | Green |
| 3rd pair | Blue – 1 |
| 4th pair | Blue – 2 |
| Outer layer | |
| 5th pair | Red |
| 6th pair | Green |
| 7th pair | Blue – 1 |
| 8th pair | Blue – 2 |
| 9th pair | Blue – 3 |
| 10th pair | Blue – 4 |
| 11th pair | Blue – 5 |
| 12th pair | Blue – 6 |
| 13th pair | Blue – 7 |
| 14th pair | Blue – 8 |
| 15th pair | Blue – 9 |

REQUIREMENTS:

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20 °C to +80 °C.

Flammability. Insulation flammability testing shall be performed in accordance with the VW-1 flame test method of UL 2556.



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Shield integrity test.*

| Frequency | Response |
|---|---|
| 100 kHz | 115 dB or more below reference |
| 500 kHz | 107 dB or more below reference |
| 1 MHz | 101 dB or more below reference |
| 5 MHz | 87 dB or more below reference |
| 10 MHz | 81 dB or more below reference |
| 15 MHz | 77 dB or more below reference |
| 20 MHz | 73 dB or more below reference |
| 25 MHz | 69 dB or more below reference |
| 30 MHz | 65 dB or more below reference |
| 100 MHz | 45 dB or more below reference |

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.

Crosstalk test limits. All values are minimums.

| Frequency (kHz) | Near end crosstalk (dB) | Far end crosstalk (dB) |
|---|---|---|
| 40 | 57 | 57 |
| 70 | 54 | 53 |
| 100 | 52 | 51 |
| 1,000 | 53 | 51 |

Insulation resistance between shields. 100 megohms-1,000 feet minimum.

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 82 $\pm$ 5.

Part or Identifying Number (PIN). The PIN shall be M49285/02.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-49285, this document references the following: UL 2556.



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CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

Review activities:
Army – MI

(Project 6145-2021-037)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change. You should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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